

Summary of Unit-II

Data-Link layer & Media Access

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Outline

I. Introduction to Data-Link Layer

- *Link-layer Addressing*
- *Network Types*

II. Data Link Control (DLC)

- *DLC Services*
- *Data-link layer protocols*
- *HDLC*
- *PPP*

III. Media Access Control (MAC)

- *Random Access*
- *Controlled Access*
- *Channelization*

IV. Wired LANs : Ethernet

- *Standard Ethernet*
- *Fast Ethernet*
- *Gigabit Ethernet*
- *10 Gigabit Ethernet*

V. Wireless LANs

Outline

- *IEEE 802.11*
- *Bluetooth*

VI. Connecting Devices

- *Repeater*
- *Hub*
- *Switch*
- *Bridge*
- *Router*
- *Gateway*

Summary- *I. Introduction to Data-Link Layer*

- Introduced the data-link layer. Started with defining the concept of links and nodes. Described the services provided by the data-link layer. Defined two categories of links: point-to-point and broadcast links. Finally discussed two sub-layers at the data-link layer.
- Discussed link-layer addressing. Explained the rationale behind the existence of an addressing mechanism at the data-link layer. Then described three types of link-layer addresses. Discussed the Address Resolution Protocol (ARP), which maps the addresses at the network layer to addresses at the data-link layer.

Summary- *II. Data Link Control (DLC)*

- Discussed the general services provided by the DLC sublayer. Described the concept of framing and two types of frames used in this sublayer. Then discussed flow and error control. Finally, explained that a DLC protocol can be either connectionless or connection-oriented.
- Discussed some simple and common data-link protocols that are implemented at the DLC sublayer. Described the working principle of Simple Protocol and Stop-and-Wait Protocol.

Summary- *II. Data Link Control (DLC)*

- Introduced HDLC, a protocol that is the basis of all common data-link protocols in use today such as PPP and Ethernet. Discussed about configurations and transfer modes. Finally described framing and three different frame formats used in this protocol.
- Discussed the concept of PPP, a very common protocol for point-to-point access. Introduced the services provided by the protocol, its frame format and transition mode in the protocol using an FSM. Also, explained multiplexing in PPP.

Summary- *III. Media Access Control (MAC)*

- Discussed random-access protocols. Four protocols, ALOHA, CSMA, CSMA/CD, and CSMA/CA, are described in this section. These protocols are mostly used in LANs and WANs.
- Discussed controlled-access protocols. Three protocols, reservation, polling, and token-passing, are described in this section. Some of these protocol are used in LANs, but others have some historical value.
- Discussed about channelization protocols. Three protocols, FDMA, TDMA, and CDMA are described in this section. These protocols are used in cellular telephony.

Summary- *IV. Wired LANs : Ethernet*

- Discussed the Ethernet protocol in general. Explained that IEEE Project 802 defines the LLC and MAC sub-layers for all LANs including Ethernet. Also, listed the four generations of Ethernet.
- Explained the principle of Standard Ethernet with its characteristics. Discussed the addressing mechanism, which is the same in all Ethernet generations. Later discussed the access method, CSMA/CD, and reviewed the efficiency of the Standard Ethernet. Explained the encoding mechanism and the implementation of this generation.
- Described the Fast Ethernet, the second generation, which can still be seen in many places. Explained the changes in the MAC sublayer with its physical layer and the implementation of this generation.

Summary- *IV. Wired LANs : Ethernet*

- Discussed the Gigabit Ethernet, which has the rate of 1 gigabit per second. Described the MAC sublayer, physical layer and its implementation.
- Explained the concept of 10 Gigabit Ethernet. This is a new technology that can be used both for a backbone LAN or as a MAN (metropolitan area network).

Summary- *V. Wireless LANs*

- Introduced the general issues behind wireless LANs and compared the wired and wireless networks. Described the characteristics of the wireless networks and the way access is controlled in these types of networks.
- Discussed a wireless LAN defined by the IEEE 802.11 Project, defined the architecture of this type of LAN and described the MAC sublayer.
- Discussed the Bluetooth technology as a personal area network (PAN). Described the architecture of the network, the addressing mechanism, and the packet format. Different layers used in this protocol are also briefly described and compared with the ones in the other wired and wireless LANs.

Summary- *VI. Connecting Devices*

- Discussed various connecting devices. Initially described hubs and their features followed by link-layer switches (or simply switches, as they are called), and showed how they can create loops if they connect LANs with broadcast domains.

Self Assessment Questions

1. Communication with respect to data-link layer is

- a) Process-to-Process
- b) Node-to-Node
- c) Message-to-Message
- d) none of the listed

Self Assessment Questions

2. Select the input bit-string, if a bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110 and the output bit-string after stuffing is 01111100101111100101,

- a) 011111010011110101
- b) 011111010111110101
- c) 011111110111110101
- d) 011111111111110101

Self Assessment Questions

3. Services offered by data link layer are

- i. Framing ii. Error control iii. Flow control
- iv. Channel coding v. Encryption vi. Decryption
- a) i, iv, v
- b) ii, iii, vi
- c) i, ii, iii
- d) ii, iv, vi

Self Assessment Questions

4. In ARP request packet, identify the destination hardware address
- a) All 1's
 - b) All 0's
 - c) Second digit even
 - d) Second digit odd

Thank You